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Quiz VPN

1. Instalar OpenVPN en servidor MV1 y cliente MV2

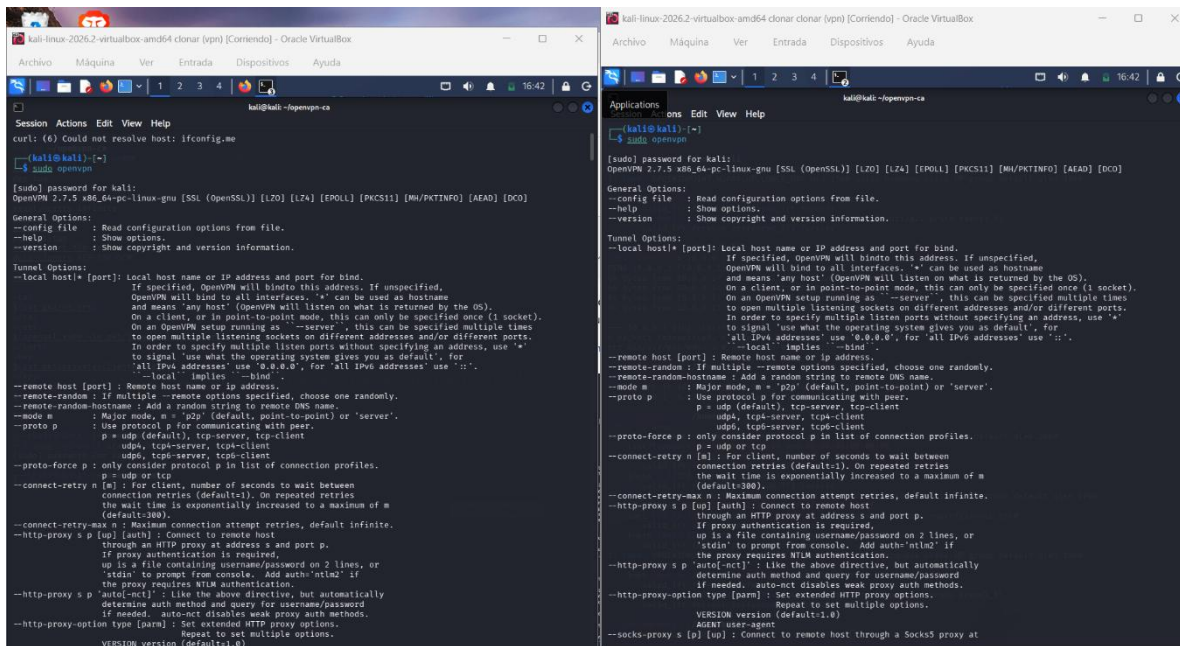


Figura 1. OpenVPN 2.7.5 en ambas VM (servidor a la izquierda, cliente a la derecha). `sudo openvpn` sin parámetros solo muestra la ayuda.

Servidor VM .6

1. Crear la CA y los certificados

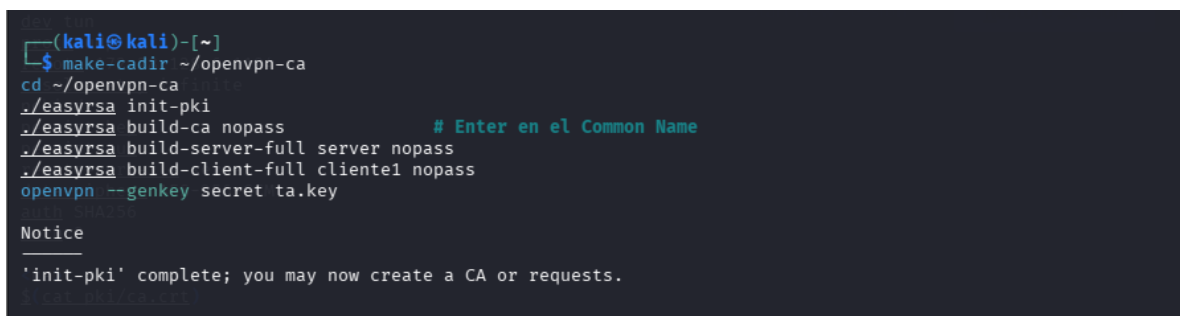
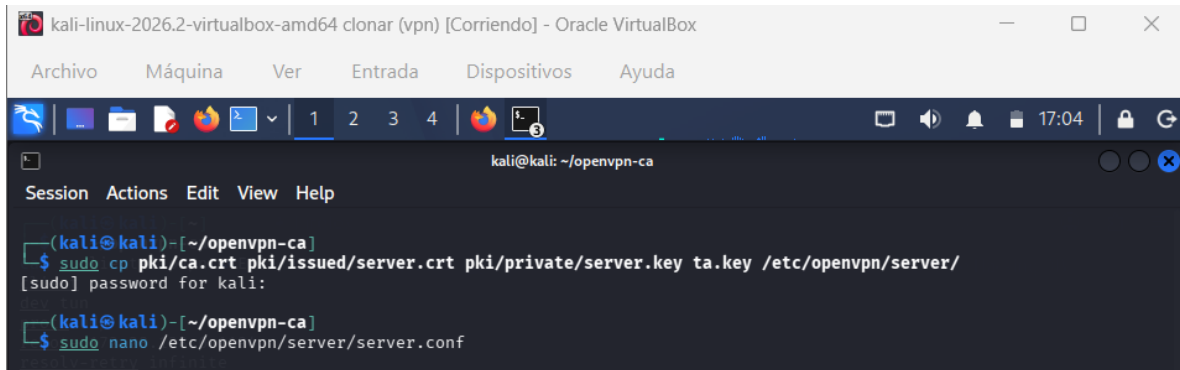


Figura 3. Creación del directorio easy-rsa e inicialización de la PKI (`init-pki complete`).

Copiar los archivos a la carpeta del servidor

Crear la configuración del servidor



```
kali@kali: ~/openvpn-ca
$ sudo cp pki/ca.crt pki/issued/server.crt pki/private/server.key ta.key /etc/openvpn/server/
[sudo] password for kali:
$ sudo nano /etc/openvpn/server/server.conf
```

Figura 4. Copia de `ca.crt`, `server.crt`, `server.key` y `ta.key` a `/etc/openvpn/server/` y edición de `server.conf`.

port 1194

proto udp

dev tun

ca ca.crt

cert server.crt

key server.key

dh none

tls-crypt ta.key

topology subnet

server 10.8.0.0 255.255.255.0

data-ciphers AES-256-GCM

auth SHA256

keepalive 10 120

persist-key

persist-tun

verb 3

Arrancar el servidor:

```
(kali㉿kali)-[~/openvpn-ca]
$ cd /etc/openvpn/server
$ sudo openvpn --config server.conf
2026-09-18 16:27:45 DEPRECATED OPTION: --persist-key option ignored. Keys are now always persisted across restarts.

2026-09-18 16:27:45 Note: --cipher is not set. OpenVPN versions before 2.5 defaulted to BF-CBC as fallback when cipher negotiation failed in this case. If you need this fallback please add '--data-ciphers-fallback BF-CBC' to your configuration and/or add BF-CBC to --data-ciphers. E.g. --data-ciphers AES-256-GCM:BF-CBC
2026-09-18 16:27:45 OpenVPN 2.7.5 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
2026-09-18 16:27:45 library versions: OpenSSL 3.6.3 9 Jun 2026, LZO 2.10
2026-09-18 16:27:45 DCO version: 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05)
2026-09-18 16:27:45 net_route_v4_best_gw query: dst 0.0.0.0
2026-09-18 16:27:45 net_route_v4_best_gw result: via 0.0.0.0 dev
2026-09-18 16:27:45 net_iface_new: add tun0 type ovpn
2026-09-18 16:27:45 DCO device tun0 opened
2026-09-18 16:27:45 ovpn-dco device [tun0] opened
2026-09-18 16:27:45 net_iface_mtu_set: mtu 1500 for tun0
2026-09-18 16:27:45 net_iface_up: set tun0 up
2026-09-18 16:27:45 net_addr_v4_add: 10.8.0.1/24 dev tun0
2026-09-18 16:27:45 Could not determine IPv4/IPv6 protocol. Using AF_INET
2026-09-18 16:27:45 Socket Buffers: R=[212992→212992] S=[212992→212992]
2026-09-18 16:27:45 UDPv4 link local (bound): [AF_INET][undef]:1194
2026-09-18 16:27:45 UDPv4 link remote: [AF_UNSPEC]
2026-09-18 16:27:45 MULTI: multi_init called, r=4096 v=4096
2026-09-18 16:27:45 IFCONFIG POOL IPv4: base=10.8.0.2 size=253
2026-09-18 16:27:45 MULTI IO: MULTI_IO INIT maxclients=1024 maxevents=1029
2026-09-18 16:27:45 Initialization Sequence Completed
2026-09-18 16:30:12 udp4:172.28.128.7:34948 VERIFY OK: depth=1, CN=Easy-RSA CA
2026-09-18 16:30:12 udp4:172.28.128.7:34948 VERIFY OK: depth=0, CN=cliente1
2026-09-18 16:30:12 udp4:172.28.128.7:34948 peer info: IV_VER=2.7.5
2026-09-18 16:30:12 udp4:172.28.128.7:34948 peer info: IV_PLAT=linux
2026-09-18 16:30:12 udp4:172.28.128.7:34948 peer info: IV_TCPNL=1
2026-09-18 16:30:12 udp4:172.28.128.7:34948 peer info: IV_MTU=1600
2026-09-18 16:30:12 udp4:172.28.128.7:34948 peer info: IV_CIPHERS=AES-256-GCM
2026-09-18 16:30:12 udp4:172.28.128.7:34948 peer info: IV_PROTO=2974
2026-09-18 16:30:12 udp4:172.28.128.7:34948 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
2026-09-18 16:30:12 udp4:172.28.128.7:34948 TLS: tls_multi_process: initial untrusted session promoted to trusted
2026-09-18 16:30:12 udp4:172.28.128.7:34948 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, signature: RSA-SHA256, peer temporary key: 768 bits X25519MLKEM768, peer signing digest/type: rsa_pss_rsae_sha256 RSASSA-PSS, key agreement: X25519MLKEM768
2026-09-18 16:30:12 udp4:172.28.128.7:34948 [cliente1] Peer Connection Initiated with [AF_INET]172.28.128.7:34948
2026-09-18 16:30:12 cliente1/udp4:172.28.128.7:34948 MULTI_sva: pool returned IPv4=10.8.0.2, IPv6=(Not enabled)
2026-09-18 16:30:12 cliente1/udp4:172.28.128.7:34948 MULTI: Learn: 10.8.0.2 → cliente1/udp4:172.28.128.7:34948
```

Figura 5. Servidor iniciado: tun0 con 10.8.0.1/24 y Initialization Sequence Completed. Después el cliente 172.28.128.7 se autentica (VERIFY OK, Peer Connection Initiated) y recibe 10.8.0.2.

Generar el archivo del cliente y habilitar SSH para copiarlo

The screenshot shows a terminal window titled "kali-linux-2026.2-virtualbox-amd64 clonar clonar (vpn) [Corriendo] - Oracle VirtualBox". The terminal is running a series of commands to copy a file and start an OpenVPN client. The first command is `scp kali@172.28.128.6:~/openvpn-ca/cliente1.ovpn ~/`. The output shows a warning about the host's authenticity and a confirmation to continue. The second command is `sudo openvpn --config ~/cliente1.ovpn`. The output shows the password for kali and a series of log messages indicating the connection process, including the use of deprecated options, cipher negotiation, and successful verification of the certificate.

```
kali@kali: ~/openvpn-ca
Session Actions Edit View Help

(kali@kali)~/openvpn-ca
$ scp kali@172.28.128.6:~/openvpn-ca/cliente1.ovpn ~/
The authenticity of host '172.28.128.6 (172.28.128.6)' can't be established.
ED25519 key fingerprint is: SHA256:thzCuvvFHYY4Uw2sXIvRmTXQ20saYfkfHKWC4nT6nE
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.28.128.6' (ED25519) to the list of known hosts.
kali@172.28.128.6's password:
cliente1.ovpn 100% 5003 2.8MB/s 00:00

(kali@kali)~/openvpn-ca
$ sudo openvpn --config ~/cliente1.ovpn
[sudo] password for kali:
2026-09-18 16:30:10 DEPRECATED OPTION: --persist-key option ignored. Keys are now always persisted across restarts.

2026-09-18 16:30:10 Note: --cipher is not set. OpenVPN versions before 2.5 defaulted to BF-CBC as fallback when cip
her negotiation failed in this case. If you need this fallback please add '--data-ciphers-fallback BF-CBC' to your
configuration and/or add BF-CBC to --data-ciphers. E.g. --data-ciphers AES-256-GCM:BF-CBC
2026-09-18 16:30:10 OpenVPN 2.7.5 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AE
AD] [DCO]
2026-09-18 16:30:10 library versions: OpenSSL 3.6.3 9 Jun 2026, LZO 2.10
2026-09-18 16:30:10 DCO version: 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05)
2026-09-18 16:30:10 TCP/UDP: Preserving recently used remote address: [AF_INET]172.28.128.6:1194
2026-09-18 16:30:10 Socket Buffers: R=[212992→212992] S=[212992→212992]
2026-09-18 16:30:10 UDPv4 link local: (not bound)
2026-09-18 16:30:10 UDPv4 link remote: [AF_INET]172.28.128.6:1194
2026-09-18 16:30:10 TLS: Initial packet from [AF_INET]172.28.128.6:1194, sid=76e59776 66dde4c7
2026-09-18 16:30:10 VERIFY OK: depth=1, CN=Easy-RSA CA
2026-09-18 16:30:10 VERIFY KU OK
2026-09-18 16:30:10 Validating certificate extended key usage
2026-09-18 16:30:10 ++ Certificate has EKU (str) TLS Web Server Authentication, expects TLS Web Server Authenticati
```

Figura 7. Copia de cliente1.ovpn por scp e inicio de la conexión en el cliente. El cliente valida el certificado del servidor con la CA (VERIFY OK).

Captura pendiente: final del log del cliente con Initialization Sequence Completed.

Verificación en el cliente

```
kali-linux-2026.2-virtualbox-amd64 clonar clonar (vpn) [Corriendo] - Oracle VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

PS> kali@kali: /home/kali

Session Actions Edit View Help

$ pwsh
PowerShell 7.6.2

(kali@kali)-[/home/kali]
PS> ip a show tun0
3: tun0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/none
    inet 10.8.0.2/24 brd 10.8.0.255 scope global tun0
        valid_lft forever preferred_lft forever
    inet6 fe80::7d70:8871:5f98:9b4b/64 scope link stable-privacy proto kernel_ll
        valid_lft forever preferred_lft forever

(kali@kali)-[/home/kali]
PS> ping -c 4 10.8.0.1
PING 10.8.0.1 (10.8.0.1) 56(84) bytes of data:
64 bytes from 10.8.0.1: icmp_seq=1 ttl=64 time=0.431 ms
64 bytes from 10.8.0.1: icmp_seq=2 ttl=64 time=0.825 ms
64 bytes from 10.8.0.1: icmp_seq=3 ttl=64 time=0.756 ms
64 bytes from 10.8.0.1: icmp_seq=4 ttl=64 time=0.585 ms

— 10.8.0.1 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3086ms
rtt min/avg/max/mdev = 0.431/0.649/0.825/0.153 ms

(kali@kali)-[/home/kali]
PS> traceroute -n 10.8.0.1
traceroute to 10.8.0.1 (10.8.0.1), 30 hops max, 60 byte packets
 1 10.8.0.1 0.592 ms 0.543 ms 0.529 ms

(kali@kali)-[/home/kali]
PS> ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:24:22:25 brd ff:ff:ff:ff:ff:ff
    inet 172.28.128.7/24 brd 172.28.128.255 scope global dynamic noprefixroute eth0
        valid_lft 462sec preferred_lft 462sec
    inet6 fe80::55b5:8cf2:2529:5210/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: tun0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/none
    inet 10.8.0.2/24 brd 10.8.0.255 scope global tun0
        valid_lft forever preferred_lft forever
    inet6 fe80::7d70:8871:5f98:9b4b/64 scope link stable-privacy proto kernel_ll
        valid_lft forever preferred_lft forever

(kali@kali)-[/home/kali]
PS> 
```

Figura 8. Cliente: tun0 con 10.8.0.2, ping a 10.8.0.1 con 0 % de pérdida (~0,65 ms) y traceroute con un solo salto.

Un solo salto (10.8.0.1) con tiempos bajos es lo esperado: la VPN une ambos extremos directamente, sin routers intermedios.

```
valid_lft forever preferred_lft forever

(kali㉿kali)-[/home/kali]
PS> ip route get 10.8.0.1
10.8.0.1 dev tun0 src 10.8.0.2 uid 1000
      cache

(kali㉿kali)-[/home/kali]
PS> █
```

```

PS> kali@kali: /home/kali

Session Actions Edit View Help

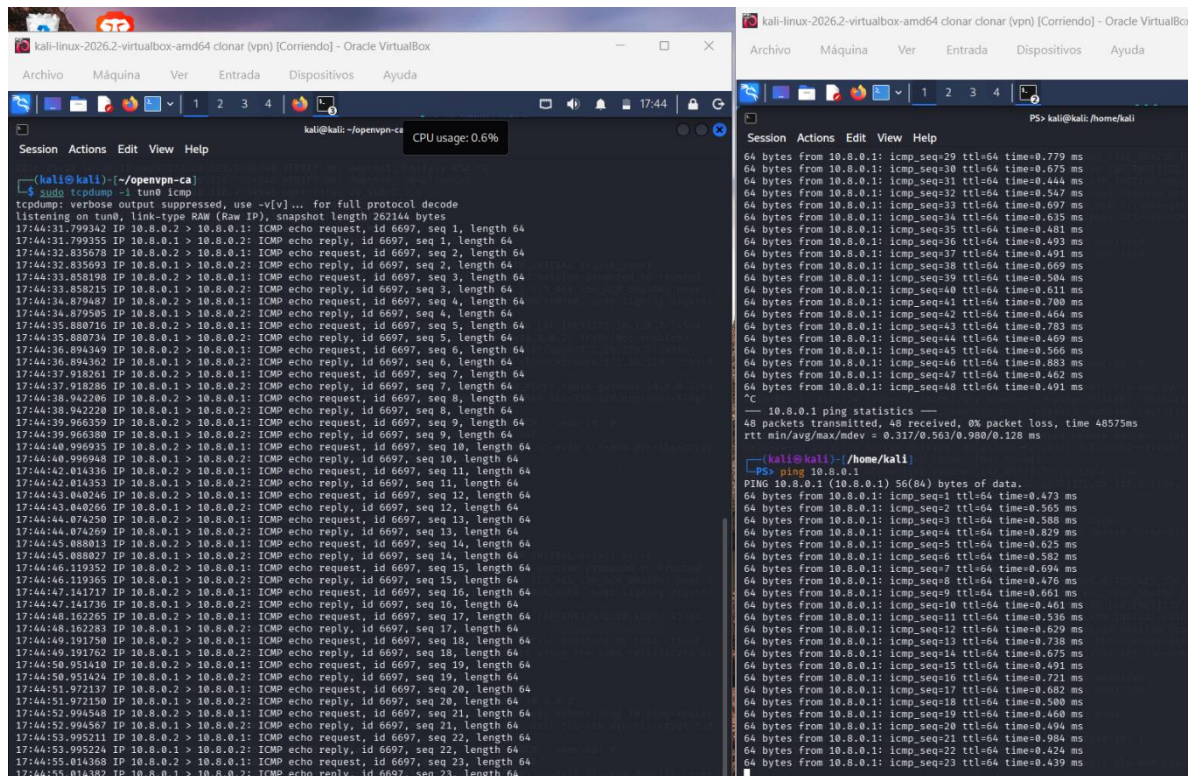
64 bytes from 172.28.128.6: icmp_seq=25 ttl=64 time=0.444 ms
64 bytes from 172.28.128.6: icmp_seq=26 ttl=64 time=0.370 ms
64 bytes from 172.28.128.6: icmp_seq=27 ttl=64 time=0.394 ms
64 bytes from 172.28.128.6: icmp_seq=28 ttl=64 time=0.386 ms
64 bytes from 172.28.128.6: icmp_seq=29 ttl=64 time=0.335 ms
64 bytes from 172.28.128.6: icmp_seq=30 ttl=64 time=0.598 ms
64 bytes from 172.28.128.6: icmp_seq=31 ttl=64 time=0.479 ms
64 bytes from 172.28.128.6: icmp_seq=32 ttl=64 time=0.387 ms
64 bytes from 172.28.128.6: icmp_seq=33 ttl=64 time=0.496 ms
64 bytes from 172.28.128.6: icmp_seq=34 ttl=64 time=0.503 ms
64 bytes from 172.28.128.6: icmp_seq=35 ttl=64 time=0.397 ms
64 bytes from 172.28.128.6: icmp_seq=36 ttl=64 time=0.589 ms
64 bytes from 172.28.128.6: icmp_seq=37 ttl=64 time=0.419 ms
64 bytes from 172.28.128.6: icmp_seq=38 ttl=64 time=0.447 ms
64 bytes from 172.28.128.6: icmp_seq=39 ttl=64 time=0.377 ms
64 bytes from 172.28.128.6: icmp_seq=40 ttl=64 time=0.414 ms
64 bytes from 172.28.128.6: icmp_seq=41 ttl=64 time=0.358 ms
64 bytes from 172.28.128.6: icmp_seq=42 ttl=64 time=0.246 ms
64 bytes from 172.28.128.6: icmp_seq=43 ttl=64 time=0.286 ms
64 bytes from 172.28.128.6: icmp_seq=44 ttl=64 time=0.339 ms
64 bytes from 172.28.128.6: icmp_seq=45 ttl=64 time=0.483 ms
64 bytes from 172.28.128.6: icmp_seq=46 ttl=64 time=0.632 ms
64 bytes from 172.28.128.6: icmp_seq=47 ttl=64 time=0.365 ms
64 bytes from 172.28.128.6: icmp_seq=48 ttl=64 time=0.519 ms
64 bytes from 172.28.128.6: icmp_seq=49 ttl=64 time=0.411 ms
64 bytes from 172.28.128.6: icmp_seq=50 ttl=64 time=0.423 ms
64 bytes from 172.28.128.6: icmp_seq=51 ttl=64 time=0.512 ms
^C64 bytes from 172.28.128.6: icmp_seq=52 ttl=64 time=0.415 ms
64 bytes from 172.28.128.6: icmp_seq=53 ttl=64 time=0.359 ms
^C

— 172.28.128.6 ping statistics —
53 packets transmitted, 53 received, 0% packet loss, time 53427ms
rtt min/avg/max/mdev = 0.246/0.428/0.632/0.081 ms

(kali@kali)-[/home/kali]
PS> ping 10.8.0.1
PING 10.8.0.1 (10.8.0.1) 56(84) bytes of data:
64 bytes from 10.8.0.1: icmp_seq=1 ttl=64 time=0.478 ms
64 bytes from 10.8.0.1: icmp_seq=2 ttl=64 time=0.554 ms
64 bytes from 10.8.0.1: icmp_seq=3 ttl=64 time=0.477 ms
64 bytes from 10.8.0.1: icmp_seq=4 ttl=64 time=0.516 ms
64 bytes from 10.8.0.1: icmp_seq=5 ttl=64 time=0.502 ms
64 bytes from 10.8.0.1: icmp_seq=6 ttl=64 time=0.621 ms
64 bytes from 10.8.0.1: icmp_seq=7 ttl=64 time=0.592 ms
64 bytes from 10.8.0.1: icmp_seq=8 ttl=64 time=0.459 ms
64 bytes from 10.8.0.1: icmp_seq=9 ttl=64 time=0.433 ms
64 bytes from 10.8.0.1: icmp_seq=10 ttl=64 time=0.572 ms
64 bytes from 10.8.0.1: icmp_seq=11 ttl=64 time=0.708 ms
64 bytes from 10.8.0.1: icmp_seq=12 ttl=64 time=0.434 ms
64 bytes from 10.8.0.1: icmp_seq=13 ttl=64 time=0.483 ms
64 bytes from 10.8.0.1: icmp_seq=14 ttl=64 time=0.625 ms

```

el ping a 172.28.128.6 (red host-only) responde con 0 % de pérdida en 53 paquetes. Después el ping a 10.8.0.1 (dentro de la VPN) también responde. Prueba que el mismo servidor es alcanzable por dos rutas distintas: la red física y el túnel.



```
kali@kali:~/openvpn-ca$ sudo tcpdump -i tun0 icmp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on tun0, link-type RAW (raw IP), snapshot length 262144 bytes
17:44:31.799342 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 1, length 64
17:44:31.799355 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 1, length 64
17:44:32.835678 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 2, length 64
17:44:32.835693 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 2, length 64
17:44:33.858198 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 3, length 64
17:44:33.858215 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 3, length 64
17:44:34.879487 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 4, length 64
17:44:34.879503 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 4, length 64
17:44:35.880716 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 5, length 64
17:44:35.880734 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 5, length 64
17:44:36.894349 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 6, length 64
17:44:36.894362 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 6, length 64
17:44:37.918261 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 7, length 64
17:44:37.918286 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 7, length 64
17:44:38.942206 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 8, length 64
17:44:38.942220 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 8, length 64
17:44:39.966350 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 9, length 64
17:44:39.966380 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 9, length 64
17:44:40.996935 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 10, length 64
17:44:40.996948 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 10, length 64
17:44:42.014336 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 11, length 64
17:44:42.014353 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 11, length 64
17:44:43.040246 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 12, length 64
17:44:43.040260 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 12, length 64
17:44:44.074290 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 13, length 64
17:44:44.074269 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 13, length 64
17:44:45.088813 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 14, length 64
17:44:45.088827 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 14, length 64
17:44:46.119352 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 15, length 64
17:44:46.119365 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 15, length 64
17:44:47.141717 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 16, length 64
17:44:47.141730 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 16, length 64
17:44:48.162283 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 17, length 64
17:44:48.162283 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 17, length 64
17:44:49.191750 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 18, length 64
17:44:49.191762 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 18, length 64
17:44:50.051410 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 19, length 64
17:44:50.051424 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 19, length 64
17:44:51.072137 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 20, length 64
17:44:51.072150 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 20, length 64
17:44:52.094548 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 21, length 64
17:44:52.094567 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 21, length 64
17:44:53.095211 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 22, length 64
17:44:53.095224 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 22, length 64
17:44:55.014360 IP 10.8.0.2 > 10.8.0.1: ICMP echo request, id 6697, seq 23, length 64
17:44:55.014382 IP 10.8.0.1 > 10.8.0.2: ICMP echo reply, id 6697, seq 23, length 64

PS> ping 10.8.0.1
PING 10.8.0.1 (10.8.0.1) 56(84) bytes of data:
64 bytes from 10.8.0.1: icmp_seq=1 ttl=64 time=0.473 ms
64 bytes from 10.8.0.1: icmp_seq=2 ttl=64 time=0.565 ms
64 bytes from 10.8.0.1: icmp_seq=3 ttl=64 time=0.588 ms
64 bytes from 10.8.0.1: icmp_seq=4 ttl=64 time=0.829 ms
64 bytes from 10.8.0.1: icmp_seq=5 ttl=64 time=0.625 ms
64 bytes from 10.8.0.1: icmp_seq=6 ttl=64 time=0.582 ms
64 bytes from 10.8.0.1: icmp_seq=7 ttl=64 time=0.694 ms
64 bytes from 10.8.0.1: icmp_seq=8 ttl=64 time=0.476 ms
64 bytes from 10.8.0.1: icmp_seq=9 ttl=64 time=0.661 ms
64 bytes from 10.8.0.1: icmp_seq=10 ttl=64 time=0.461 ms
64 bytes from 10.8.0.1: icmp_seq=11 ttl=64 time=0.536 ms
64 bytes from 10.8.0.1: icmp_seq=12 ttl=64 time=0.629 ms
64 bytes from 10.8.0.1: icmp_seq=13 ttl=64 time=0.738 ms
64 bytes from 10.8.0.1: icmp_seq=14 ttl=64 time=0.675 ms
64 bytes from 10.8.0.1: icmp_seq=15 ttl=64 time=0.491 ms
64 bytes from 10.8.0.1: icmp_seq=16 ttl=64 time=0.721 ms
64 bytes from 10.8.0.1: icmp_seq=17 ttl=64 time=0.682 ms
64 bytes from 10.8.0.1: icmp_seq=18 ttl=64 time=0.508 ms
64 bytes from 10.8.0.1: icmp_seq=19 ttl=64 time=0.468 ms
64 bytes from 10.8.0.1: icmp_seq=20 ttl=64 time=0.494 ms
64 bytes from 10.8.0.1: icmp_seq=21 ttl=64 time=0.984 ms
64 bytes from 10.8.0.1: icmp_seq=22 ttl=64 time=0.424 ms
64 bytes from 10.8.0.1: icmp_seq=23 ttl=64 time=0.439 ms
```

mientras el cliente hace ping a 10.8.0.1, el servidor ve en tun0 los ICMP echo request/reply entre 10.8.0.2 y 10.8.0.1. Prueba que el tráfico entra y sale por el túnel.

Conclusiones:

Se configuró una VPN punto a punto con OpenVPN 2.7.5 entre dos máquinas Kali en una red host-only. Se creó una CA propia con easy-rsa y se emitieron certificados para el servidor y el cliente, además de una clave `tls-crypt`. La conexión se estableció con autenticación mutua por certificados y cifrado AES-256-GCM. Se comprobó con la interfaz `tun0` en ambos extremos (10.8.0.1 y 10.8.0.2) y con un ping sin pérdida de paquetes. Con esto se confirma que el cliente se comunica con el servidor a través de un túnel cifrado.